CSE400 - Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

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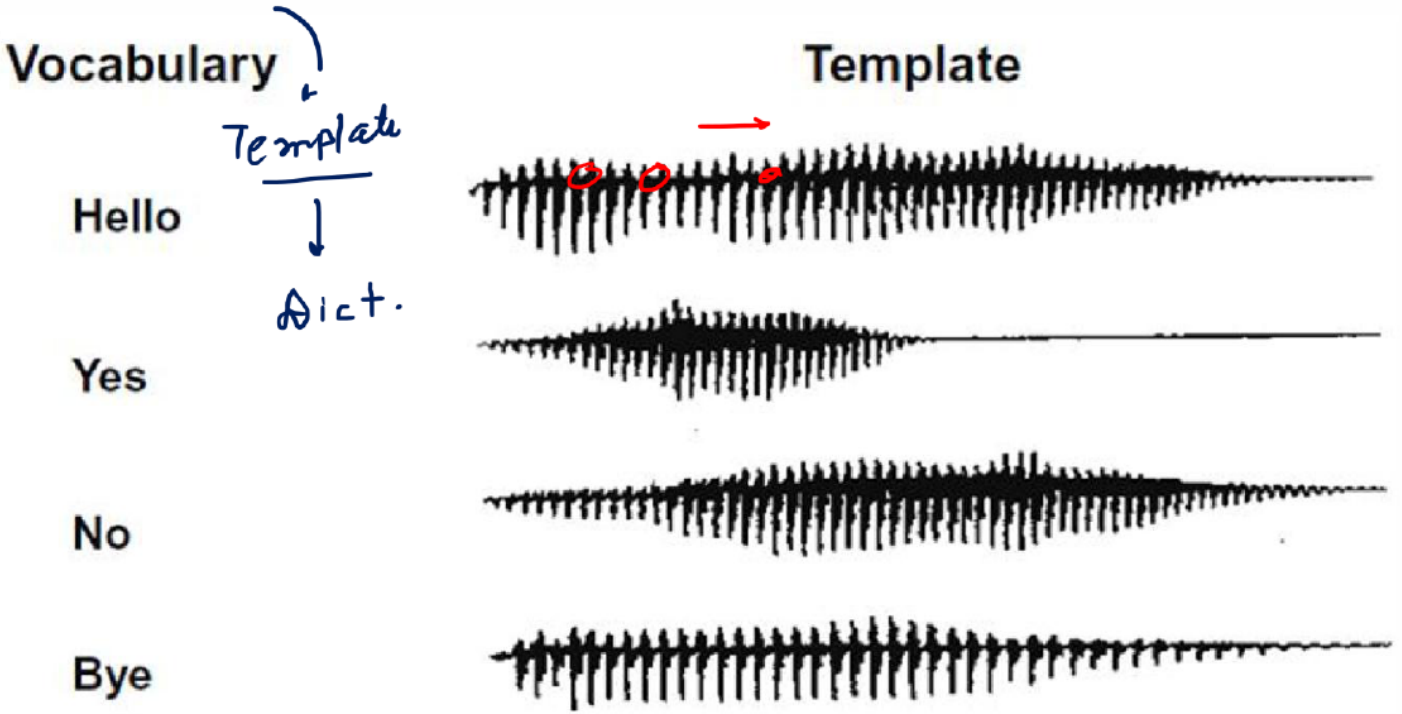
January 15, 2026

Outline-1/2

- **CSE400: General Course Information**
  - Team
  - Active Learning Platform: Campuswire
  - Schedule
  - Grading
- **Why should we learn CSE400?**
  - Example: Daily life conversations
- **Engineering Applications:**
  1. Speech Recognition
  2. System Radar System
  3. Communication Network

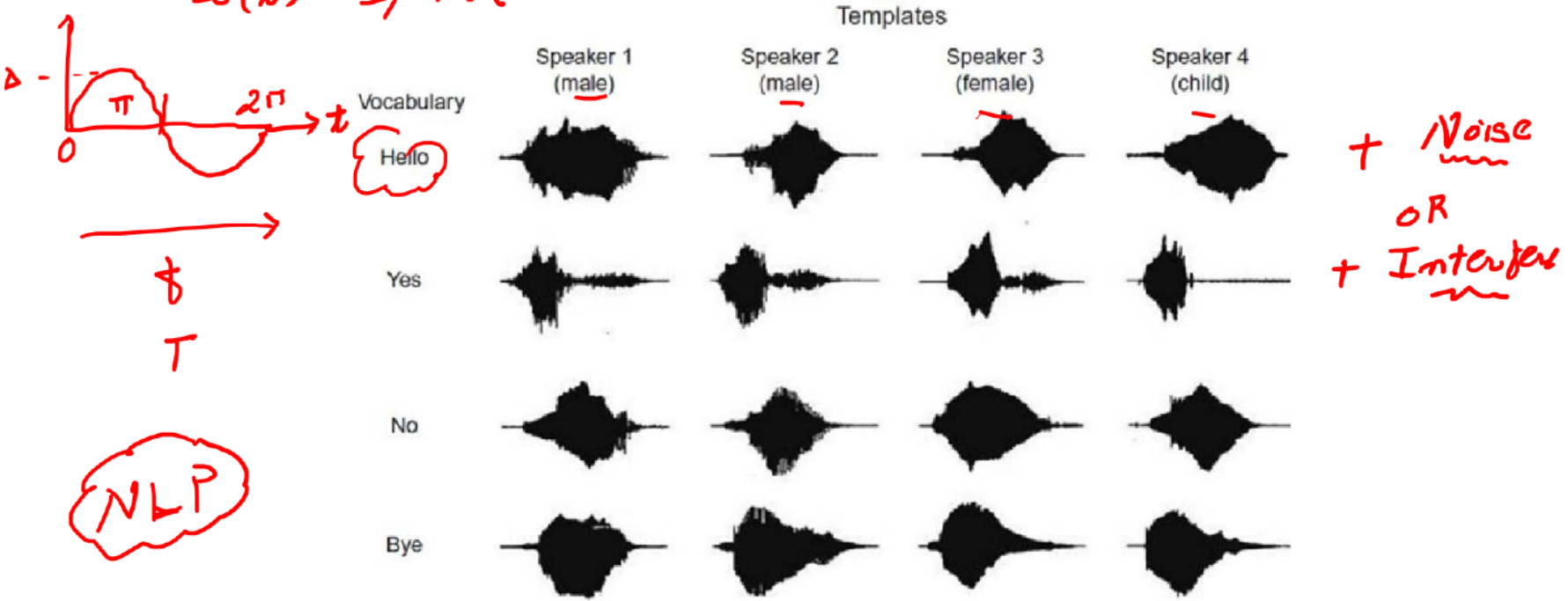
Engineering applications

1. Speech Recognition System (1/2)



Engineering applications

1. Speech Recognition System (2/2)



Engineering applications

2. Radar System

SVPA



Source: http://www.ktm2day.com/2011/01/12/tia-13-yr-old-radar-still-out-of-order/

$P_e = 10^{-7}$

$H_0 : y_i = w_i$

$H_1 : y_i = s_i + w_i$

Hardware

$w_i \sim CN(0, \sigma^2)$, μ

Tx	Rx	
0	0	✓
0	1	→ False Alarm
1	0	→ Miss Detect
1	1	✓

$\downarrow P_M$

$P_D \uparrow$

$P_M + P_D = 1$

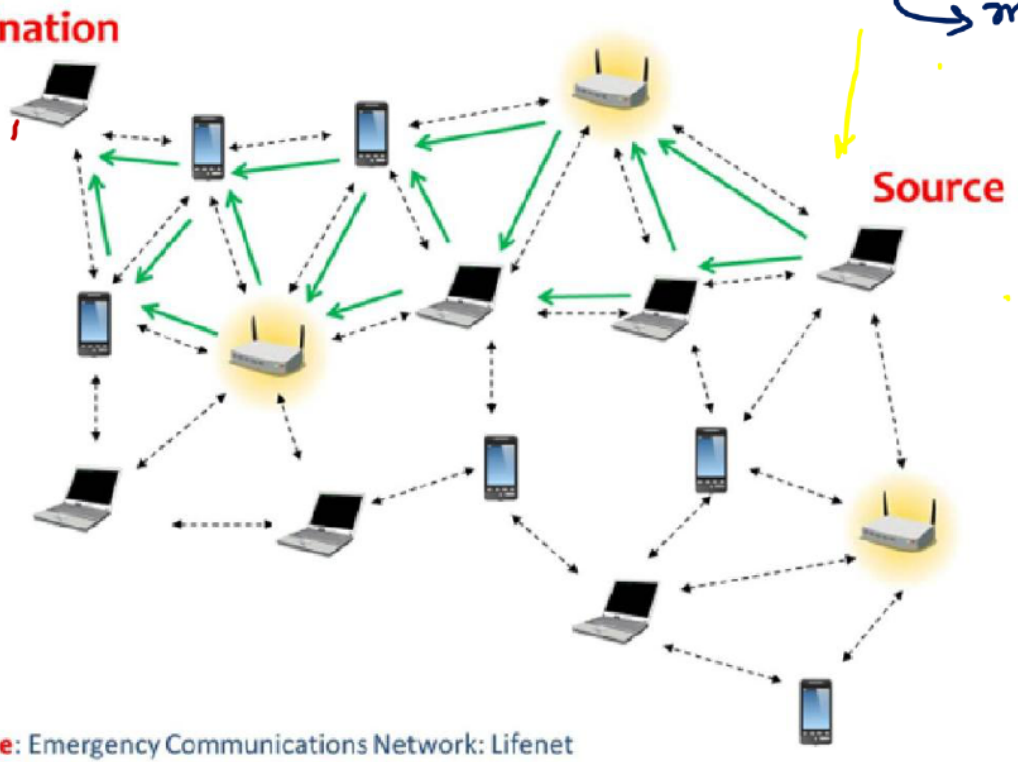
Engineering applications

3. Communication Network

- WiFi
  - IEEE 802.11 a, b, g, n ac, bb
  - 2.4 GHz
  - 5 GHz
  - 6 GHz → WiFi 6 ✗
- Cellular n/w
  - 4G, 5G
- STA
- WiFi 5

✓ Arrival of Packets

✓ QoS: Delay (Latency) → ms/sec



Source: Emergency Communications Network: Lifenet

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  - Axioms of Probability
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Introduction to Probability Theory

Experiments, Sample Space and Events

Definitions:

- **Experiment (E):** It is a procedure we perform, that produces some result.
  Example: Tossing a coin five times (E_5)

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  Example: One of the outcome of E_5 is: $\xi_1 = HHTHT$

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- **Event (Any Letter):** It is a certain set of outcomes of an experiment.
  Example: Event C with an experiment E_5
  $C = \{$all outcomes consisting of an even number of heads$\}$

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- **Event (Any Letter):** It is a certain set of outcomes of an experiment.
  Example: Event C with an experiment E_5
  $C = \{$all outcomes consisting of an even number of heads$\}$
- **Sample space (S):** It is the collection or set of "all possible" distinct (Mutually Exclusive and Collectively Exhaustive) outcomes of an experiment.
  - Example: Flip a coin
  - **Mutually Exclusive:** You can get heads or tails, but not both.
  - **Collectively Exhaustive:** You cannot get anything other than heads or tails.

Definitions:

- **Sample space (S):** [Continued]
  - S: It is the universal set of outcomes of an experiment.
  - Sample space is Discrete or Countably infinite or Continuous.

Introduction to Probability Theory

Experiments, Sample Space and Events



Definitions:

- **Sample space (S):** [Continued]
  - S: It is the universal set of outcomes of an experiment.
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Examples:

1. Flipping a fair coin once
2. Cubical die with numbered faces is rolled
3. Rolling two dices
4. Flip a coin until a tails occurs
5. Random number generator with interval [0, 1)

Introduction to Probability Theory

Axioms of Probability

- Next step: Assign probabilities to various outcomes and events
- **Probability:** It is a measure of the likelihood of various events OR It is a function of an event that produces a numerical quantity that measures the likelihood of that event.
- There are many ways to define such a function.

Axioms: The statements are taken to be self-evident and require no proof.

- For any event A, $0 \leq Pr(A) \leq 1$
- If S is the sample space for a given experiment, $Pr(S) = 1$

(a) If $A \cap B = \emptyset$, then $Pr(A \cup B) = Pr(A) + Pr(B)$

(b) For an infinite number of mutually exclusive sets, A_i, $i = 1, 2, 3...$, $A_i \cap A_j = \emptyset$ for all $i \neq j$,

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

Introduction to Probability Theory

Corollary from Probability Axioms

- Define A as the null event for all values of i greater than n.

Corollary 2.1

For M finite number of mutually exclusive sets, A_i, $i = 1, 2, 3...$, $A_i \cap A_j = \emptyset$ for all $i \neq j$,

$$Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} Pr(A_i)$$

- Axiom 3 is equivalent to Corollary 2.1 when the sample space is finite.
- However, the added generality of Axiom 3 is necessary when the sample space consists of an infinite number of points.

Introduction to Probability Theory

Propositions from Probability Axioms

Introduction to Probability Theory

Propositions from Probability Axioms

- **Proposition 2.1:** $Pr(A^c) = 1 - Pr(A)$

Introduction to Probability Theory

Propositions from Probability Axioms

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- **Proposition 2.2:** If $A \subset B$ then $Pr(A) \leq Pr(B)$

Introduction to Probability Theory

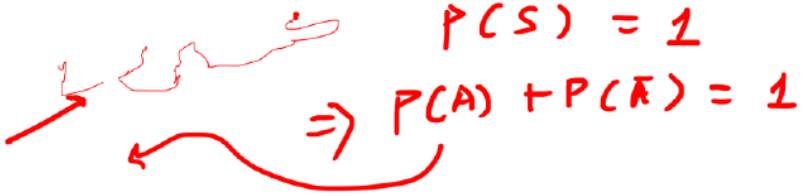
Propositions from Probability Axioms

- **Proposition 2.1:** $Pr(A^c) = 1 - Pr(A)$
- **Proposition 2.2:** If $A \subset B$ then $Pr(A) \leq Pr(B)$
- **Proposition 2.3:** For any sets and (not necessarily mutually exclusive),

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Introduction to Probability Theory

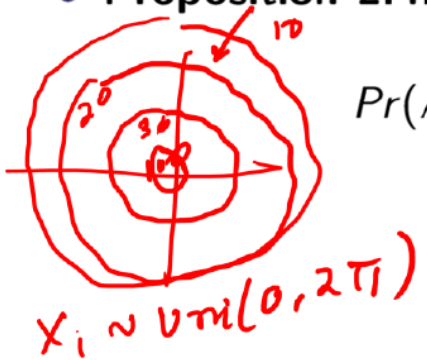
Propositions from Probability Axioms



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- **Proposition 2.4:**



$$Pr(A_1 \cup A_2 \cup \cdots \cup A_M) = \sum_{i=1}^{M} Pr(A_i) - \sum_{i_1 < i_2} Pr(A_{i_1} A_{i_2}) + \ldots$$
$$+ (-1)^{r+1} \sum_{i_1 < i_2 < \cdots < i_r} Pr(A_{i_1} A_{i_2} \ldots A_{i_r})$$
$$+ \cdots + (-1)^{M+1} Pr(A_1 A_2 \ldots A_M)$$

= 1/2 = 1/2

- **Example# 2.6:** Coin flipping experiment: Compute the $Pr(H)$ and $Pr(T)$
- **Example# 2.7:** Dice rolling experiment: Compute the Pr(even number is rolled) {2, 4, 6}
- **Example# 2.8:** A pair of dice: Compute the $Pr(A)$, where A is the event of the sum of two dice numbers equal to five. {(2,3) (3,2) (4,1) (1,4)}

1) Example: A pair of dice. Compute $Pr(A)$, where A is the event of sum of two dice number equal to five.

Simulation of Dice Tossing Experiment:

n	1000	2000	3000	4000	5000	6000	7000	8000	9000	10,000
n_A	96	200	314	408	521	630	751	859	970	1095
n_A/n	0.096	0.100	0.105	0.102	0.104	0.105	0.107	0.107	0.108	0.110

2) Inference:

- To get an exact measure of the probability of an event, we must be able to repeat the event an infinite number of times [a serious drawback to this approach].
- Many random phenomena in which we might be interested are not repeatable.

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Motivation, Notation, and Concepts of Joint Probability

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- **Calculation Approaches:**
  1. **Classical:**
     - Both events A and B can be expressed in terms of atomic outcomes.
     - We then write as the set of those atomic outcomes that are common to both.
     - Calculate the probabilities of each of these outcomes.

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     - Both events A and B can be expressed in terms of atomic outcomes.
     - We then write as the set of those atomic outcomes that are common to both.
     - Calculate the probabilities of each of these outcomes.
  2. **Relative Frequency:** Let $n_{A,B}$ be the number of times A and B occur simultaneously in n trials:

$$Pr(A, B) = \lim_{n \to \infty} \frac{n_{A,B}}{n}$$

Joint Probability

Example 1: Card Deck Example

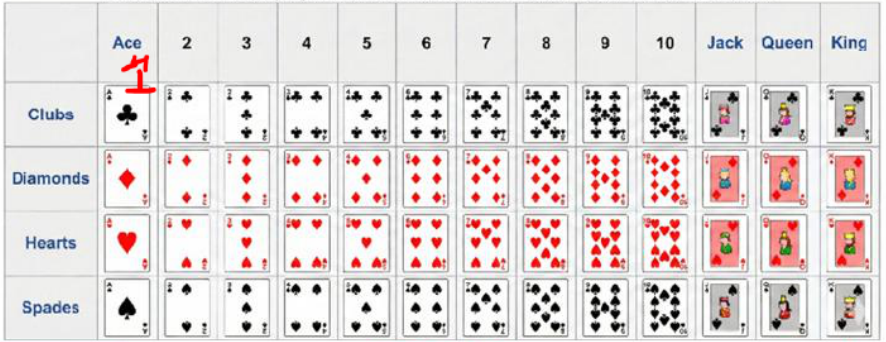
Card Deck Example

Consider a set of 52 playing cards. Let:

- $A = \{\text{Red card selected}\}$
- $B = \{\text{Number card selected}\}$ (Ace is a number card)
- $C = \{\text{Heart card selected}\}$

Find: $Pr(A)$, $Pr(B)$, $Pr(C)$, $Pr(A, B)$, $Pr(A, C)$, $Pr(B, C)$

Example set of 52 playing cards; 13 of each suit clubs, diamonds, hearts, and spades



Joint Probability

Example 1: Card Deck Example

Solution:

- **Individual Probabilities:**
  - $Pr(A) = \frac{26}{52} = \frac{1}{2}$ (26 red cards: 13 hearts + 13 diamonds)
  - $Pr(B) = \frac{40}{52} = \frac{5}{26}$ (10 number cards in each of 4 suits)
  - $Pr(C) = \frac{13}{52} = \frac{1}{4}$ (13 hearts in deck)
- **Joint Probabilities:**
  - $Pr(A, B) = \frac{20}{52} = \frac{5}{26}$ (10 red number cards in hearts + 10 in diamonds)
  - $Pr(A, C) = \frac{13}{52} = \frac{1}{4}$ (all 13 hearts are red)
  - $Pr(B, C) = \frac{10}{52}$ (10 number cards that are hearts)

Joint Probability

Example 2: Costume Party Example

The Scenario

Alex has limited clothing options left:

- 3 food-stained white t-shirts
- 1 radioactive green cape
- 2 pairs of old flannel dinosaur pajama pants
- 4 pairs of polka-dot boxer shorts

If Alex selects one top and one bottom randomly, what is the probability her outfit is the **radioactively-green cape** paired with the **bright polka-dot boxers**?

Options:

- A) 1/2
- B) 1/4
- C) 1/6

Joint Probability

Example 2: Costume Party Example

- **Step 1: Analyze Tops**
  - Total Tops = 3 (T-shirts) + 1 (Cape) = 4.
  - $Pr(\text{Cape}) = 1/4$.

Joint Probability

Example 2: Costume Party Example

- **Step 1: Analyze Tops**
  - Total Tops = 3 (T-shirts) + 1 (Cape) = 4.
  - $Pr(\text{Cape}) = 1/4$.
- **Step 2: Analyze Bottoms**
  - Total Bottoms = 2 (Pants) + 4 (Boxers) = 6.
  - $Pr(\text{Boxers}) = 4/6 = 2/3$.

Joint Probability

Example 2: Costume Party Example

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- **Step 2: Analyze Bottoms**
  - Total Bottoms = 2 (Pants) + 4 (Boxers) = 6.
  - $Pr(\text{Boxers}) = 4/6 = 2/3$.
- **Step 3: Joint Probability (Independence)**
  - Total Probability = $Pr(\text{Cape}) \times Pr(\text{Boxers})$
  - Calculation: $(1/4) \times (4/6) = 1/6$.

Correct Option: C

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- **Definition:** The probability of A conditioned on knowing B occurred:

$$Pr(A|B) = \frac{Pr(A,B)}{Pr(B)} \quad \text{where } Pr(B) > 0$$

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- **The conditional probability of three events:**

$$Pr(A,B,C) = Pr(C|A,B)Pr(A,B) = Pr(C|A,B)Pr(B|A)Pr(A)$$

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- **The conditional probability of three events:**

$$Pr(A,B,C) = Pr(C|A,B)Pr(A,B) = Pr(C|A,B)Pr(B|A)Pr(A)$$

- **Conditional probability of M events (Chain Rule):**

$$Pr(A_1, A_2, \ldots A_M) = Pr(A_M|A_1, A_2, \ldots A_{M-1}) \times Pr(A_{M-1}|A_1, A_2, \ldots A_{M-2}) \cdots Pr(A_2|A_1) \cdot Pr(A_1)$$

Conditional Probability

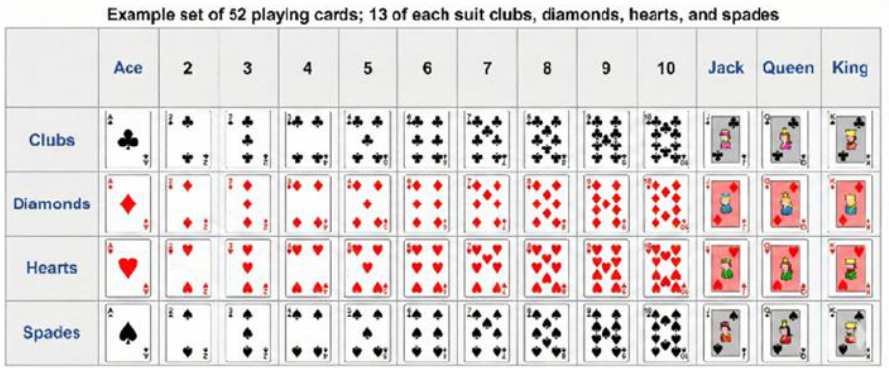
Example 3: Cards Without Replacement

Problem Statement

We select two cards at random. When we select the second card, we **do not return** the first card to the deck.

Find: $Pr(B|A)$, where:

- A: First card was a Spade
- B: Second card was a Spade



Conditional Probability

Example 3: Cards Without Replacement

- **Initial State:** A standard deck has 52 cards, 13 of which are Spades
- **Event A Occurs:** One Spade is removed.

Conditional Probability

Example 3: Cards Without Replacement

- **Initial State:** A standard deck has 52 cards, 13 of which are Spades
- **Event A Occurs:** One Spade is removed.
- **Remaining State:**
  - Total cards remaining $= 52 - 1 = 51$.
  - Spades remaining $= 13 - 1 = 12$.

Conditional Probability

Example 3: Cards Without Replacement

- **Initial State:** A standard deck has 52 cards, 13 of which are Spades
- **Event A Occurs:** One Spade is removed.
- **Remaining State:**
  - Total cards remaining $= 52 - 1 = 51$.
  - Spades remaining $= 13 - 1 = 12$.
- **Calculation:**

$$Pr(B|A) = \frac{\text{Remaining Spades}}{\text{Remaining Total}} = \frac{12}{51}$$

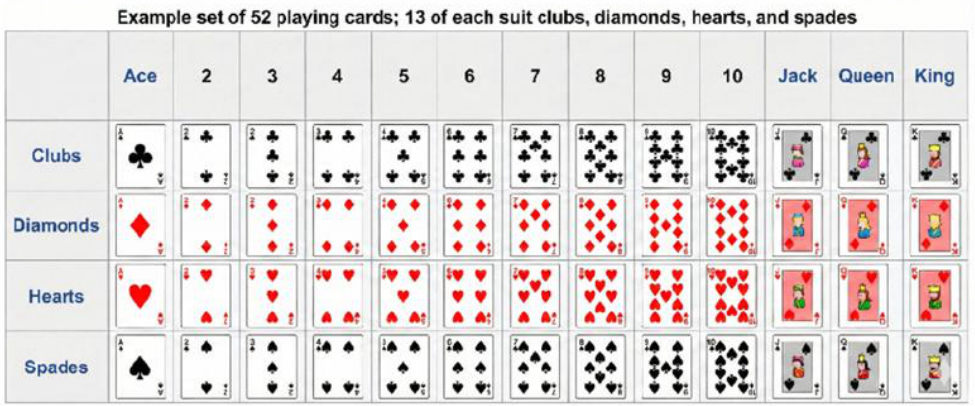
Conditional Probability

Example 4: Game of Poker

Flush Probability

You are dealt 5 cards. A **flush** is when you are dealt all five cards of the same suit.

1. What is the probability you are dealt a flush in Spades?
2. What is the probability of a flush in *any* suit?



Hint: Start with defining event $A_i = i^{th}$ card dealt to us is a spade.

Conditional Probability

Example 4: Game of Poker

- **Part 1: Flush in Spades**
  - Prob 1st is Spade: 13/52
  - Prob 2nd is Spade: 12/51 ... and so on.
  - $P(\text{Spade Flush}) = \frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} \times \frac{10}{49} \times \frac{9}{48}$

Conditional Probability

Example 4: Game of Poker

- **Part 1: Flush in Spades**
  - Prob 1st is Spade: 13/52
  - Prob 2nd is Spade: 12/51 ... and so on.
  - $P(\text{Spade Flush}) = \frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} \times \frac{10}{49} \times \frac{9}{48}$
- **Part 2: Any Flush**
  - There are 4 suits (Spades, Hearts, Diamonds, Clubs).
  - Since events are mutually exclusive:

$$P(\text{Any Flush}) = 4 \times P(\text{Spade Flush})$$

Conditional Probability

Example 5: The Missing Key

The Scenario

Bob is 80% certain her key is in her jacket (40% Left Pocket, 40% Right Pocket).
If a search of the **left-hand pocket** does not find the key, what is the conditional probability that it is in the other pocket?

Options:

- A) 2/5
- B) 1/3
- C) 2/3

Conditional Probability

Example 5: The Missing Key

- **Define Events:**
  - L: Key is in Left Pocket ($P(L) = 0.4$)
  - R: Key is in Right Pocket ($P(R) = 0.4$)
  - K: Key is in the jacket ($P(K) = 0.8$)
- **Goal:** Find $P(R|L^c)$ (Prob. in Right given NOT in Left).

Conditional Probability

Example 5: The Missing Key

- **Define Events:**
  - L: Key is in Left Pocket ($P(L) = 0.4$)
  - R: Key is in Right Pocket ($P(R) = 0.4$)
  - K: Key is in the jacket ($P(K) = 0.8$)
- **Goal:** Find $P(R|L^c)$ (Prob. in Right given NOT in Left).
- **Calculation:**

$$P(R|L^c) = \frac{P(R \cap L^c)}{P(L^c)}$$

Since R implies L^c (key can't be in both), $P(R \cap L^c) = P(R)$.

$$= \frac{P(R)}{1 - P(L)} = \frac{0.4}{1 - 0.4} = \frac{0.4}{0.6} = \frac{2}{3}$$

Correct Option: C

Open Campuswire For Class Participation.